EVERY LOW-TWO RECURSIVELY ENUMERABLE SET IS ALMOST-EVERYWHERE WEAK-GENERIC COMPUTABLE

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Editorial note. Complete original author mathematical body, both main priority constructions and embedded bibliography retained; one main theorem per task. Internal verification statements are supporting material rather than additional counted units. No mathematical alteration, proof repair, added assumption or independent correctness certification.

Allowed standard prerequisites (editorial).

- Computability theory, relative weak genericity and effective measure. Native AEWG, superhigh/low-two, box/claim objects and all author priority construction definitions and verification arguments retained.. The complete main construction proof is retained across its full section, rather than treating its short final proof paragraph as the whole argument.

1. Introduction

The main results in this paper are pure r.e. degree theoretical statements. However, the motivation comes from coarse computability, a new area that has received attention of mathematical logicians in the last two decades. We say that two sets $A, B \subseteq \omega$ are *coarsely similar*, if their symmetric difference has asymptotic density 0. Coarse similarity is an equivalence relation on $\mathcal{P}(\omega)$. Let $\mathcal{S}$ denote the set of all coarse similarity classes and write $[A]$ for the coarse similarity class of the set A.

Hirschfeldt, Jockusch and Schupp [4] investigated the natural metric $\delta([A],[B])$ on $\mathcal{S}$, which is equal to the upper asymptotic density of $\{m : A(m) \neq B(m)\}$. They then defined a metric H between Turing degrees as follows: Given a Turing degree $\mathbf{d}$, a set $A \subseteq \omega$ is *coarsely* $\mathbf{d}$*-computable at density* $r \in [0,1]$ if there is a set $C \leq_T \mathbf{d}$ such that the lower asymptotic density of the set $\{m : A(m) = C(m)\}$ is no less than r. The *coarse* $\mathbf{d}$*-computability bound* of a set A is the supremum of all reals r for which A is coarsely $\mathbf{d}$-computable at density r. Let $\bar{\mathbf{d}} = \{[A] : \gamma_{\mathbf{d}}(A) = 1\}$ for any degree $\mathbf{d}$. The *Hausdorff distance*, $H(\mathbf{d},\mathbf{e})$, between the degrees $\mathbf{d}$ and $\mathbf{e}$ is the Hausdorff distance of $\mathbf{d}$ and $\mathbf{e}$ in $\mathcal{S}$.

It was shown in [4] that H is a $\left\{0, \frac{1}{2}, 1\right\}$-valued metric between degrees, which is a relativization of Monin's Theorem [6]. A degree $\mathbf{a}$ is *dispersive* if $H(\mathbf{a}, \deg(B)) = 1$ for almost every set B, where $\deg(B)$ is the Turing degree of B. A degree $\mathbf{a}$ is *attractive* if $H(\mathbf{a}, \deg(B)) = \frac{1}{2}$ for almost every set B. The Kolmogorov 0-1 Law (see [3]) implies that any measurable Turing invariant set (a collection of sets that is closed under Turing equivalence) has measure 0 or 1. In particular, every Turing degree is either dispersive or attractive. Certain classes of degrees are proved to be contained in the dispersive or attractive degrees.

Theorem 1 ([4])**.** *(1) There is a dispersive high r.e. degree.*
(2) Every low r.e. degree is dispersive.
(3) The class of dispersive degrees is closed downwards.
(4) Every almost everywhere dominating degree is attractive.

A set $W \subseteq 2^{<\omega}$ is *dense* if for every string σ, there is a string $\tau \in W$ extending σ. Given $n \geq 1$, a set A is *weakly n-generic* if for every dense Σ^0_n set $W \subseteq 2^{<\omega}$ there is m such that $A \upharpoonright m \in W$ (sometimes we write $A \in [W]$). The definition can be relativized to any oracle. For example, a set A is *weakly 1-Z-generic* if for every dense $\Sigma^0_1(Z)$ set $W \subseteq 2^{<\omega}$ there is m such that $A \upharpoonright m \in W$.

Definition 2. *A set B is* AEWG *if almost every set computes a set that is weakly 1-B-generic. A degree $\mathbf{b}$ is AEWG if there is an AEWG set $B \in \mathbf{b}$.*

It was pointed out in [4] that for a degree, being AEWG implies being dispersive. But it is still unknown whether being dispersive implies being AEWG even within the r.e. degrees.

Question 3 ([4, part of Question 9.2])**.** *Can a set A be dispersive without it being AEWG?*

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In fact, Theorem 1(1) and (2) were derived as a corollary from the following theorem.

Theorem 4 ([4]). *There is an AEWG high r.e. set and every low r.e. set A is AEWG.*

Dobrinen and Simpson [2] defined a degree $\mathbf{a}$ to be *almost everywhere dominating (a.e.d.)* if for almost every set B, every B-recursive function f is dominated by an $\mathbf{a}$-recursive function g, that is $(\exists m)(\forall n > m)[f(n) < g(n)]$. A set A is *superhigh* if $A' \geq_{\mathrm{tt}} \emptyset''$. Simpson [9] showed that every a.e.d degree is superhigh and there exists an r.e. set which is superhigh and not a.e.d. See Nies [8] page 361-362 for a detailed diagram of the implications of some upward (downward) closed properties for Turing degrees and r.e. degrees including the properties mentioned above.

The following question was asked in [4]: Do the notions of being attractive and being a.e.d. coincide for r.e. degrees? Recently, Hirschfeldt and Royer [5] showed that if a degree is not diagonally non-recursive, then it is attractive iff it is a.e.d. and hence answered the question in the affirmative. However, their proof does not use the AEWG property, but rather works directly with Monin's notion of 2^{2^n}-infinitely often equality. It is still unknown whether being dispersive implies being AEWG for an r.e. degree. So the following question is the next step towards Question 3: *If A is an r.e. set which is not a.e.d., must A be AEWG?* In view of this, we obtain the following results.

Theorem 5. *There is an AEWG superhigh r.e. set.*

Theorem 6. *Every low$_2$ r.e. set is AEWG.*

The two proofs of are very different in the way of arranging the tree of strategies.

Ng [7] constructed an r.e. minimal pair of superhigh degrees (independently by Shore (unpublished)). We follow the framework in [7] for constructing a superhigh r.e. set in Theorem 5 in Section 2.

In the proof of Theorem 6 in Section 3 we use the "certification process" developed recently by Cholak, Downey and Greenberg [1].

We assume the familiarity with recursion theory as in [10] and early chapters in [3], for instance. We adopt the convention of using uppercase Greek letters for functionals, and lowercase Greek letters for their use. We append $[s]$ to functionals or their use to describe their values at stage s.

2. Proof of Theorem 5

2.1. **Requirements.** To show that the r.e. set A to be constructed is superhigh, it is enough to build a truth-table reduction $A' \geq_{\mathrm{tt}} \mathrm{Tot} := \{e : \Phi_e \text{ is total}\}$. We divide this reduction into positive requirements P_e: $e \in \mathrm{Tot} \iff A' \vDash \sigma_e$, where the truth tables σ_e are defined uniformly in the construction. We also build a functional Ψ and meet the negative requirements

$$Q_e : \quad W_e^A \text{ dense } \Rightarrow (\forall X)\big(\Psi^X \text{ total } \rightarrow (\exists n)\, \Psi^X \upharpoonright n \in W_e^A\big).$$

Finally we meet the global requirement M: the set of oracles X such that Ψ^X is total has positive measure. If all Q_e and M are satisfied, then on a positive-measure class the value Ψ^X is weakly 1-A-generic. The class of oracles computing a weakly 1-A-generic is Turing invariant, so the Kolmogorov 0-1 law raises this positive measure to measure one.

2.2. **The basic module.** The usual basic module for Q_e as in [4] fixes a parameter $k = k_e$ and tries, one by one, cones $[\rho]$ with $|\rho| = k$. While a cone is claimed, the construction keeps Ψ constant along that cone until all Ψ-values of current leaves of the cone have extensions in W_e^A; at that point it extends Ψ into W_e^A on the whole claimed piece and moves to the next cone.

The point of the present version is that a higher-priority node may already claim some proper subcones of $[\rho]$. If these subcones do not cover $[\rho]$, the node for Q_e claims the clopen remainder. Thus a claim is not just a string, but a pair (ρ, F), where $|\rho| = k$, F is a finite prefix-free set of proper extensions of ρ, and $R(\rho, F) := [\rho] \setminus \bigcup_{\xi \in F} [\xi] \neq \emptyset$. We call ρ the *base* of the claim and $R(\rho, F)$ the *claimed clopen set*. At a later stage s, the module waits until every string τ of length $s-1$ with $[\tau] \subseteq R(\rho, F)$ has an extension of Ψ^τ in W_e^A. It then defines $\Psi^{\tau 0} = \Psi^{\tau 1}$ to be such an extension, for all these τ, and drops the claim. The excluded subcones are left to the higher-priority nodes which already claimed them. If there is no such a stage s, we may believe that W_e^A is not dense. And if we carefully define the parameters k_e, the waste of measure in the case that there is no such a stage s is tame, so we meet requirement M.

2.3. **The tree.** As usual, our construction is performed on a tree T of strategies. Nodes on T represent guesses about the eventual behaviour of some aspects of the construction, and use their guesses to meet a requirement they are assigned to. We will define T to be a finitely branching recursive tree (but not the binary tree). The outcomes at odd levels are ∞ and f, ordered $\infty <_{\mathrm{left}} f$. The outcomes at even levels are finite initial segments of ω, ordered in the usual way. We use the notation $\alpha <_{\mathrm{left}} \beta$ and $\alpha <_L \beta$ in

the standard sense: $\alpha <_L \beta$ means that α is either strictly to the left of β or is a proper initial segment of β. Nodes of length $2e+1$ are *odd nodes* and are assigned to P_e. Each odd node γ has two *boxes* (see subsection 2.4), denoted by η^γ_∞ and η^γ_f. Nodes of length $2e$ are *even nodes* and are assigned to Q_e. Each even node α has a *parameter* $k_\alpha > 1$ and at any stage has at most one current claim (ρ, F) and $|\rho| = k_\alpha$. A node α is said to be an *R-node*, if it is assigned to the requirement R.

The parameters and even outcomes are defined recursively as follows. Put $k_\lambda = 2$. Suppose the parameter has been defined for all even nodes of length at most $2n$, and let $\sigma_0, \ldots, \sigma_{m-1}$ be the nodes of length $2n$. The outcomes of σ_i are $0, 1, 2, \ldots, 2^{k_{\sigma_i}+1}+1$, where even outcomes correspond to expansionary action and odd outcomes to non-expansionary waiting. This defines level $2n+1$. The nodes of length $2n+2$ have the form σx, where $|\sigma| = 2n+1$ and $x \in \{\infty, f\}$. List them from left to right as $\tau_0, \ldots, \tau_j$. Choose the least $c > 2$ which is not the parameter of any shorter even node and set $k_{\tau_i} = c + i$.

At each stage s we define an approximation to the true path, δ_s of length s. A node α is *visited* at stage s, if $\alpha \subseteq \delta_s$. The leftmost path on T such that every initial segment is visited infinitely often is the *true path*. Since T is finitely branching, the true path exists. At each stage s we will define Ψ^τ for all $\tau \in 2^s$.

2.4. **Boxes and the truth table.** The concepts in this subsection originate from Ng [7]. We define an auxiliary functional Θ in the construction. Fix a partial recursive 1-1 function $(\gamma, x) \mapsto \eta^\gamma_x$ defined for γ odd and $x \in \{\infty, f\}$ such that $\Theta^A(\gamma, x) \downarrow$ if and only if $A'(\eta^\gamma_x) = 1$. We will not use the function $(\gamma, x) \mapsto \eta^\gamma_x$ in the construction. η^γ_x is just the name of a box by the follows.

Definition 7. *Given s, an odd $\gamma \in T$ and $x \in \{\infty, f\}$, we say that at stage s:*

(i) The box η^γ_x is empty *if $\Theta^A(\gamma, x)[s] \uparrow$.*
(ii) To fill *the empty box η^γ_x with use u means that we define $\Theta^{A\restriction u+1}(\gamma, x)[s] \downarrow$ with use u.*
(iii) To clear *the box η^γ_x which has already been filled with use u at some stage $t < s$ such that $A_t \restriction u+1 = A_s \restriction u+1$ means that we enumerate u into A. To clear an empty box means doing nothing.*

Now fix e and list the nodes of length $2e+1$ from left to right as $\gamma_0 <_{\text{left}} \cdots <_{\text{left}} \gamma_n$. Put $\rho_i = \gamma_{n-i}$. The truth table σ_e reads this level in the order $\eta^{\rho_0}_f, \eta^{\rho_0}_\infty, \eta^{\rho_1}_f, \eta^{\rho_1}_\infty, \ldots, \eta^{\rho_n}_f, \eta^{\rho_n}_\infty$. It ignores everything after the first occupied box. Its value is 1 exactly when this first occupied box is an η_∞-box. Equivalently,

$$\sigma_e = \bigvee_{i \le n} \left(A'(\eta^{\rho_i}_\infty) \wedge \neg A'(\eta^{\rho_i}_f) \wedge \bigwedge_{j<i} \left(\neg A'(\eta^{\rho_j}_f) \wedge \neg A'(\eta^{\rho_j}_\infty) \right) \right).$$

Since the tree is recursive and each level is finite, the sequence $\{\sigma_e\}_{e\in\omega}$ is recursive.

2.5. **Legal residual claims and expansionary stages.** At a stage, if an even node β has a current claim, let $b(\beta)$ denote its base. For an even node α and a string ρ of length k_α, let

$$H_s(\alpha) = \{b(\beta) : \beta <_L \alpha \text{ and } \beta \text{ has a current claim at stage } s\}.$$

We say that ρ is *legal for α at stage s* if no member of $H_s(\alpha)$ is an initial segment of ρ, and if the prefix-minimal members of $\{\xi \in H_s(\alpha) : \rho \subset \xi\}$ do not cover $[\rho]$. If ρ is legal, let $F_s(\alpha, \rho)$ be this finite prefix-minimal set. Then $R_s(\alpha, \rho) = [\rho] \setminus \bigcup_{\xi \in F_s(\alpha,\rho)} [\xi]$ is the clopen remainder which α may claim. When α claims ρ, it records the pair $(\rho, F_s(\alpha, \rho))$; the recorded set F is not changed until the claim is dropped or α is initialized. If $C = R(\rho, F)$ is a current claim, say that τ is *inside* C if $[\tau] \subseteq C$.

Definition 8. *Each even node α carries a* restraint *r_α, initialized to -1. After the last initialization of α, r_α is the maximum of the oracle uses of all W^A_e-enumerations which have been used by α in successful extensions of Ψ.*

Definition 9. *Let α be an even node of length $2e$. A stage $s > 0$ is* α-expansionary *if:*

(i) α is visited at stage s;
(ii) α has a current claim $C = R(\rho, F)$ and $s - 1 \ge \max\{|\xi| : \xi \in F \cup \{\rho\}\}$;
(iii) for every $\tau \in 2^{s-1}$ inside C, there is a string in $W^A_e[s]$ properly extending Ψ^τ. Among all stage-s computations witnessing such an extension, effectively choose one of least oracle use, and denote the extension by ν_τ;
(iv) if w is the maximum of the oracle uses of the enumerations chosen in (iii), and $w^+ = \max\{w, r_\alpha\}$, then
  (a) for every odd β with $\beta\infty \subseteq \alpha$, the box η^β_f is empty or has use greater than w^+;
  (b) for every odd $\gamma >_{\text{left}} \alpha$, both η^γ_f and η^γ_∞ are empty or have use greater than w^+.

Definition 10. *For an odd node β of length $2e+1$, define $l(\beta,s) < s$ to be the maximal y such that $\Phi_e(x)[s]\downarrow$ for all $x < y$. A stage $s > 0$ is* β-expansionary *if β is visited at s and $l(\beta,s) > l(\beta,s^-)$, where $s^- < s$ is the last stage at which β was visited. If s is the first stage at which β is visited, we assume $l(\beta,s^-) = 0$ by convention.*

If $l(\beta,s)$ constantly increases, then we have reason to believe that Φ_e is total.

The key device in the construction is an *outcome jump* at even nodes. After the last initialization of an even node α, its visits are organized in adjacent pairs $c, c+1$, where c is even: c is the expansionary outcome, while $c+1$ is the corresponding waiting outcome. Suppose that, at an α-expansionary stage, the computations which α is about to preserve have use bound w^+, but a box already occupied at an odd proper extension of α has use at most w^+. Instead of returning to outcome c, the construction advances to the next even outcome $c+2$; relative to the immediately preceding outcome this is a jump of two from c, or of one from $c+1$. Every such box lies on an earlier branch whose α-outcome is below $c+2$, and the jump therefore places it strictly to the left of the new path approximation. Unless a node to the left of α is later visited or α is later initialized, all subsequent outcomes at α are at least $c+2$, so this box can no longer be cleared by either end-of-stage clearing operation. Any box filled after the jump receives a fresh use greater than w^+ and is harmless to the protected computations. Thus the jump disposes of every existing small-use threat; the only possible exceptions are precisely the higher-priority injuries excluded once the true-path node has stabilized.

2.6. **The construction.** At stage 0, define $\Psi^\lambda = \lambda$ and regard every even node as *initialized* for convenience (to initialize an even node we mean we delete its claim if it has a current claim, declare it to be not satisfied if it is currently satisfied, and set its restraint r_α to -1).

At stage $s > 0$, we inductively define δ_s of length s and let the initial segments of δ_s to act. Suppose we have defined $\alpha = \delta_s \upharpoonright 2e$ for $2e < s$. We allow the even nodes to act as soon as we see that it is on δ_s.

Case 1. α was not initialized earlier at this stage, is not satisfied, and s is α-expansionary. Let $C = R(\rho, F)$ and w^+ be as in Definition 9.

Subcase 1-1. There is no odd $\gamma \supset \alpha$ and $x \in \{\infty, f\}$ such that the box η^γ_x has use at most w^+. Proceed as follows.

(1) For every $\tau \in 2^{s-1}$ inside C, choose ν_τ as in Definition 9 and define $\Psi^{\tau 0} = \Psi^{\tau 1} = \nu_\tau$. After these extensions have been made, set the restraint $r_\alpha = w^+$.
(2) Drop the claim C. Then let α claim the $<_{\text{left}}$-least string ρ' of length k_α which is legal for α currently and has not been used as a base by α since its last initialization. If there is no such ρ', declare α is *satisfied*.
(3) Initialize all even $\beta \supset \alpha$.
(4) Let $c < 2^{k_\alpha+1}+1$ be the even number such that, at the last visit to α since its last initialization (such a stage exists since α has a claim currently), the outcome of α was c or $c+1$. Define $\delta_s(2e) = c$.

Subcase 1-2. Some odd $\gamma \supset \alpha$ has a box η^γ_x of use at most w^+. Perform steps (1)-(3) from Subcase 1-1. Let c be as in Subcase 1-1(4) and define $\delta_s(2e) = c+2$. This is always possible since α can have at most 2^{k_α} expansionary actions between two initializations.

Case 2. α was not initialized earlier at this stage, is not satisfied, has a current claim, and s is not α-expansionary. Let c be as in Subcase 1-1(4) and define $\delta_s(2e) = c+1$.

Case 3. α was not initialized earlier at this stage, is not satisfied, and has no current claim (this could happen if s is the first stage at which α is visited since the last stage at which α was initialized). Let α claim the $<_{\text{left}}$-least string ρ of length k_α which is legal for α at this stage and has not been used as a base by α since the last initialization of α. If no such ρ exists, declare α satisfied. Initialize all even $\beta \supset \alpha$, and define $\delta_s(2e) = 1$ if a claim was made and $\delta_s(2e) = 0$ otherwise.

Case 4. α was not initialized earlier at this stage and is satisfied. If α became satisfied in Case 3 without making a claim, define the outcome to be 0. Otherwise let c be as in Subcase 1-1(4) and define $\delta_s(2e) = c$.

Case 5. α was initialized earlier at this stage. Define $\delta_s(2e) = 1$.

We say that an even node *acts* at stage s if Case 1 or Case 3 holds.

Now suppose $\beta = \delta_s \upharpoonright (2e+1)$ has been defined and $2e+1 < s$. Define $\delta_s(2e+1) = \infty$ if s is β-expansionary, and define $\delta_s(2e+1) = f$ otherwise. After δ_s of length s has been defined, perform the following end-of-stage actions.

(i) Initialize all even $\beta >_{\text{left}} \delta_s$.
(ii) For every odd $\gamma >_{\text{left}} \delta_s$, clear η^γ_∞ and η^γ_f.
(iii) For every odd α such that $\alpha\infty \subseteq \delta_s$, clear η^α_f.

(iv) For every odd $\alpha \subset \delta_s$, in increasing order of length, fill $\eta^{\alpha}_{\delta_s(|\alpha|)}$ if it is empty, with the use a fresh number larger than every number which has appeared so far.
(v) For each $\mu \in 2^s$ for which Ψ^μ has not yet been defined, if $[\mu]$ is contained in one of the current claimed clopen sets, define $\Psi^\mu = \Psi^{\mu\restriction(s-1)}$. Otherwise define $\Psi^\mu = \Psi^{\mu\restriction(s-1)}0$.

2.7. **Verification.**

Lemma 11. *At the end of every stage, the bases of distinct current claims have pairwise distinct lengths, and the clopen sets currently claimed by even nodes are pairwise disjoint.*

Proof. The recursive assignment gives distinct parameters to distinct even nodes. Since a node only uses bases of length equal to its parameter, the base lengths of distinct current claims are pairwise distinct. For disjointness, consider the stage at which an even node α makes a new claim with base ρ and exclusion set F. Any node extending α is initialized immediately, and any node to the right of the final approximation is initialized at the end of the stage. Thus it is enough to compare the new claim with claims of nodes $\beta <_L \alpha$.

Let $\eta = b(\beta)$. If η is an initial segment of ρ, then ρ is not legal, so this case cannot occur. If ρ is an initial segment of η, then either η is itself a member of F, or it extends a prefix-minimal member of F; in either case the new claimed clopen set excludes $[\eta]$ and hence is disjoint from the claim of β. If ρ and η are incomparable, the corresponding cylinders are already disjoint. □

Lemma 12. *Suppose that at the end of a stage a box attached to an odd node γ is nonempty. If ρ is an odd initial segment of γ, and $\rho x \subseteq \gamma$ for $x \in \{\infty, f\}$, then the box η^ρ_x is nonempty and its use is smaller than the use of the box at γ.*

Proof. Boxes are filled only after all clearings for the stage have been performed, and they are filled along the current approximation from top to bottom with fresh uses. Hence, at the stage when a box at γ is filled, each relevant predecessor box is already nonempty and has smaller use. If later an enumeration into A destroys the predecessor computation, then it also destroys the computation at γ, whose use is larger. Thus the relation holds at every stage. □

Lemma 13. *Let ρ be an even node on the true path. Then ρ is initialized only finitely many times, and after its last initialization it acts only finitely many times.*

Proof. The proof is by induction on the length of ρ. Once no node to the left of ρ is visited, only proper initial segments of ρ can initialize it; by induction, those nodes act only finitely often.

Assume now that ρ is on the true path and work after its last initialization. No higher-priority node changes its claim after this point. Hence the set of legal bases of length k_ρ is fixed. The node ρ never uses the same base twice between two initializations, and there are only 2^{k_ρ} possible bases. Case 3 can occur only when ρ has no current claim, and each later Case 1 drops one claim. Hence ρ acts only finitely many times. □

Lemma 14. *Let α be an even node and suppose that, after the last initialization of α, Subcase 1-2 is applied to α at a stage s. Let w^+ be the corresponding bound, and suppose that the outcome assigned to α at stage s is $c+2$. Assume further that after stage s no node to the left of α is visited and α is not initialized. Then no box attached to an odd proper extension of α which is nonempty at the beginning of the end-of-stage actions at stage s and has use at most w^+ is ever cleared after stage s.*

Proof. Let η^γ_x be a box as in the statement. Since it is already nonempty before the end-of-stage actions at stage s, it was filled at an earlier stage at which the approximation to the path passed through γ. Then the outcome of α on that earlier path was strictly smaller than $c+2$. Hence, once stage s assigns the outcome $c+2$ to α, the node γ is strictly to the left of the current approximation below α. At every later stage at which α is visited, the outcome of α is again at least $c+2$, unless α has first been initialized; the latter is excluded by hypothesis. Thus γ remains strictly to the left of the later approximations whenever the construction passes through α. The ordinary end-of-stage clearing of boxes strictly to the right of the current approximation therefore never clears η^γ_x. The clearing of an f-box along an ∞-outcome also cannot clear it, since that operation only applies to odd nodes which are initial segments of the current approximation. Finally, a clearing caused by an even proper initial segment of α would initialize α, again contrary to the hypothesis. Therefore the box is never cleared after stage s. □

Lemma 15. *Let α be the Q_e-node on the true path. Let s_0 be a stage after which the left of α is never visited and α is never initialized. Suppose that α drops a claim $C = R(\rho, F)$ at an α-expansionary stage $s > s_0$. Then for all $X \in C$, if Ψ^X is total, then $(\exists n)\Psi^X \restriction n \in W^A_e$.*

Proof. Let s be the stage at which the claim is dropped. For every $\tau \in 2^{s-1}$ inside C, the construction chooses $\nu_\tau \in W_e^A[s]$ extending Ψ^τ and defines $\Psi^{\tau 0} = \Psi^{\tau 1} = \nu_\tau$. Let w be the maximum of the oracle uses of these finitely many enumerations. At the same step of the construction we set $r_\alpha = w^+ = \max\{w, r_\alpha[s]\}$. Hence every W_e^A-enumeration used by α since its last initialization has oracle use at most r_α after stage s.

It remains to show that no later enumeration into A below w^+ can occur. The protection clause in the definition of α-expansionary, applied with w^+, ensures that at stage s there is no dangerous box of use at most w^+ above α along an ∞-outcome, nor to the right of α. Thus later clearings of boxes of these two kinds cannot change A below w^+: boxes already present there have use greater than w^+, and boxes filled later have fresh use greater than w^+. It remains to consider boxes attached to odd extensions of α. In Subcase 1-1 there is no such extension box of use at most w^+. In Subcase 1-2, the only possible small extension boxes are boxes already present at stage s. By Lemma 14, these boxes are not cleared later while the hypotheses of the present lemma remain in force. □

Lemma 16. *All negative requirements are satisfied.*

Proof. Suppose, towards a contradiction, that some Q_e fails. Then W_e^A is dense, while for some X, Ψ^X is total and $\Psi^X \upharpoonright n \notin W_e^A$ for all n. Let α be the Q_e-node on the true path. Fix a stage s_0 after which α is never initialized and no node to its left is visited. We may take s_0 to be the stage of the last initialization of α: any later visit to a node to its left, or any later change in a higher-priority claim relevant to α, would initialize α again. Thus all such claims are fixed after s_0, and every base used by α after this initialization is claimed at a stage greater than s_0.

We first show that α is eventually satisfied. By Lemma 13, choose $s_1 > s_0$ after the last action of α. Suppose that α is not satisfied. Since α is visited infinitely often, it must then have a claim $C = R(\rho, F)$ which remains current after s_1; otherwise its next visit would fall under Case 3 and make α act. Choose $t \geq \max\{|\xi| : \xi \in F \cup \{\rho\}\}$. For each $\tau \in 2^t$ inside C, density gives a string in W_e^A properly extending Ψ^τ. Choose stable computations witnessing these finitely many extensions, let u bound their oracle uses, and put $u^+ = \max\{u, r_\alpha\}$. While C remains current, clause (v) keeps Ψ constant above each such τ inside C. By the least-use convention in Definition 9(iii), at every sufficiently late stage the witnesses chosen there therefore have use at most u. Choose the first visit to α at a stage s so late that the stable computations and the number u^+ have appeared, $s - 1 \geq t$, and the bound in the preceding paragraph applies. At this visit the stage-specific value of w^+ is at most u^+. If the visit is expansionary, then Case 1 already makes α act. Otherwise only the protection condition can fail, and the end-of-stage actions clear all boxes relevant to Definition 9(iv) whose uses are at most u^+. Any such box filled afterwards has fresh use greater than u^+. Hence the next visit to α is expansionary, and Case 1 again makes α act. Either case contradicts the choice of s_1. Thus α is eventually satisfied.

Now suppose that the base of a fixed higher-priority claim is an initial segment of X, and choose such a base η of maximal length; there are only finitely many fixed claims. If the corresponding claim is $R(\eta, F)$, then no member of F is an initial segment of X, since its recorded exclusions are bases of fixed higher-priority claims and any such member would be longer than η. Consequently $X \in R(\eta, F)$. This claim remains current, so clause (v) keeps $\Psi^{X\upharpoonright m}$ constant for all sufficiently large m, contrary to the totality of Ψ^X. Therefore no fixed higher-priority base is an initial segment of X. Let $\rho = X \upharpoonright k_\alpha$. The preceding conclusion implies that ρ is legal for α. Indeed, no fixed higher-priority base is an initial segment of ρ, and the fixed bases extending ρ cannot cover $[\rho]$, since X belongs to none of their cones. Moreover, X belongs to the corresponding clopen remainder. Since the higher-priority claims are fixed, ρ remains legal after s_0. After its last initialization, α uses each legal base at most once and is declared satisfied only when no legal unused base remains. Thus ρ was claimed and its claim was later dropped after s_0. Lemma 15 yields $(\exists n)\, \Psi^X \upharpoonright n \in W_e^A$, contrary to the choice of X. □

Lemma 17. *The global requirement M is satisfied.*

Proof. For each stage s, let C_s be the union of the clopen sets currently claimed by even nodes at the end of stage s. If $X \notin C_s$ for some arbitrarily large stages s, then clause (v) appends a bit to Ψ along X at arbitrarily large stages, and hence Ψ^X is total. Therefore the set of oracles X such that Ψ^X is not total is a subset of $\liminf_s C_s$. By Lemma 11, the current claimed clopen sets are pairwise disjoint. If α has base length k_α, then its claimed clopen set is contained in a cone of measure 2^{-k_α}. The parameters k_α are pairwise distinct and all at least 2, so for every s, $\mu(C_s) \leq \sum_{n\geq 2} 2^{-n} = \frac{1}{2}$. Fatou's lemma for measure gives $\mu(\liminf_s C_s) \leq \liminf_s \mu(C_s) \leq \frac{1}{2}$. Thus Ψ is total on a class of measure at least $1/2$. □

Lemma 18. *All positive requirements are satisfied.*

Proof. Fix e. We verify that the truth table σ_e satisfies $e \in \mathrm{Tot} \iff A' \vDash \sigma_e$. Let α be the node of length $2e+1$ on the true path. Choose a stage s_0 after which no node to the left of α is visited and no even proper initial segment of α acts. From then on, every node of length $2e+1$ strictly to the right of α has both boxes cleared whenever α is visited. Boxes to the left of α may remain occupied, but the truth table reads the level from right to left, so they are irrelevant once one of the boxes of α is permanent.

Suppose first that Φ_e is total. Then α has infinitely many expansionary stages. Choose an α-expansionary stage $s_1 > s_0$. At the end of s_1, the box η_f^α has been cleared and η_∞^α is filled if it was empty. Let u be the use of η_∞^α. We check that no later enumeration into A below u can destroy this computation. No node strictly to the left of α is visited after s_0, and no even proper initial segment of α acts after s_0. Every box to the right of α, and every f-box on an odd predecessor whose ∞-outcome leads to α, has just been cleared or will be refilled only with fresh use greater than u. Every box properly below $\alpha\infty$ which is nonempty after s_1 has use greater than u by Lemma 12, and boxes filled later also have fresh use greater than u. Therefore later clearings can enumerate only numbers greater than u. Hence η_∞^α is permanently occupied, while η_f^α is cleared at later α-expansionary stages and is not permanent. Therefore $A' \vDash \sigma_e$.

Suppose now that Φ_e is not total. Then α has only finitely many expansionary stages. Choose $s_1 > s_0$ after the last such stage and then a later visit s_2 to α with outcome f, after all boxes to the right of α filled before s_2 have been cleared. At the end of s_2, the box η_f^α is filled if it is empty; let u be its use. There is no later expansionary visit to α, so the construction never again clears η_f^α because of an ∞-outcome at α. Also, after s_0 there is no left visit and no action by an even proper initial segment of α. The boxes to the right of α which existed before s_2 have already been cleared, any box below the later path has use greater than u by Lemma 12, and every box filled after s_2 receives fresh use greater than u. Thus every later clearing which can occur has use greater than u, and η_f^α is permanently occupied. Since the truth table reads η_f^α before η_∞^α, we have $A' \nvDash \sigma_e$. □

Proof of Theorem 5. By Lemma 18, A is superhigh. By Lemma 16, 17 and the discussion in Subsection 2.1, A is AEWG. □

3. Proof of Theorem 6

Fix a low_2 r.e. set A and a recursive increasing enumeration $A = \bigcup_s A_s$. Since A is given to us, we have only the requirements Q_e and M from Subsection 2.1.

In the previous section, when we properly extend the definition of Ψ, the negative requirement imposes its restraint by clear boxes in order to keep the enumerations in W_e^A. Now we need to consider the case that some number less than the use of enumerations will be enumerated into A later which we cannot control. To coexist with that, we use the "certification process" developed recently by Cholak, Downey and Greenberg [1]. The authors used this technique together with the Δ_3 guessing argument to prove that every low_2 r.e. set has an atomless hyperhypersimple superset, which is a milestone towards the Soare's low_2 Conjecture. This result is irrelevant but the technique in their proof is significant for this section.

The proof that every low r.e. set has the AEWG property in [4] uses Robinson's trick for guessing. This is done by asking $\Sigma_1^0(A)$ questions to $\emptyset'$, and using the approximation solutions to the questions given by the Limit Lemma and the Recursion Theorem to decide what to do in the construction. We do not use guessing arguments here though there is a version of guessing called the Δ_3 guessing which is suitable for low_2 r.e. sets (see [1] for example). Instead, the certification process uses domination property of low_2 sets and there is no need for the Recursion Theorem. We will explain how we use the "certification process" after introducing the setup.

We first record the consequence of the relativized High Domination Theorem that will be used in the certification process.

Lemma 19. *There is a $\emptyset'$-recursive function f which dominates every total A-recursive function. Moreover, f has a recursive approximation $\{f_s\}_{s\in\omega}$ such that $f_s(x) \le f_{s+1}(x)$ for every x.*

Proof. Since A is r.e., $A \le_T \emptyset'$. Since it is low_2, $A'' \equiv_T \emptyset'' = (\emptyset')'$. Hence $\emptyset'$ is high relative to A. The relativized High Domination Theorem gives an $\emptyset'$-recursive function h which dominates every total A-recursive function. Let $\{h_s\}$ be a convergent recursive approximation to h and put $f_s(x) = \max_{t\le s} h_t(x)$, $f(x) = \lim_s f_s(x)$. For each fixed x the limit exists. Also $f \ge h$, so f still has the required domination property, and its displayed approximation is monotone. □

The monotonicity of $f_s(x)$ will imply that, at each fixed coordinate, only finitely many distinct attempt records ever become certified.

3.1. **The priority tree and objects.** We first define all of the finite objects used by the stage construction. As numerous concepts are defined in this section, every concept will be typeset **in bold** at its first appearance.

The tree of strategies is $T = \omega^{<\omega}$. A node α of length e runs the module for Q_e, and its children are $\beta_i = \alpha i$ $(i \in \omega)$, which are workers for Q_e. As in Section 2, write $\gamma <_{\text{left}} \beta$ when γ lies lexicographically to the left of β, and put $\gamma <_L \beta$ if $\gamma \subsetneq \beta$ or $\gamma <_{\text{left}} \beta$.

Fix a recursive one-to-one parameter function $k : T \setminus \{\lambda\} \longrightarrow \{2, 3, 4, \ldots\}$, $k(\beta) = k_\beta$, such that $k_{\alpha i} < k_{\alpha j}$ whenever $i < j$. We call k_β the **parameter** of β.

As in Subsection 2.2, a **claim** for a node β is a pair (ρ, F), where $\rho \in 2^{k_\beta}$, F is a finite prefix-free set of proper extensions of ρ, and $R(\rho, F) = [\rho] \setminus \bigcup_{\xi \in F} [\xi] \neq \varnothing$. We call ρ the **base** of the claim and $R(\rho, F)$ its **claimed clopen set**. At any stage a nonempty node has at most one current claim. If the current claim of γ is (ρ, F), put $C_s(\gamma) = R(\rho, F)$; if γ has no current claim, put $C_s(\gamma) = \varnothing$. For a worker β, define its **higher-priority claim set** by $\mathcal{H}_s(\beta) = \bigcup_{\gamma <_L \beta} C_s(\gamma)$. For $\rho \in 2^{k_\beta}$, the **current residual** of the base ρ for β is $R_s(\beta, \rho) = [\rho] \setminus \mathcal{H}_s(\beta)$. When this residual is nonempty, let $F_s(\beta, \rho)$ be the finite prefix-minimal frontier for $\mathcal{H}_s(\beta) \cap [\rho]$. Then every member of $F_s(\beta, \rho)$ is an extension of ρ and $R_s(\beta, \rho) = R(\rho, F_s(\beta, \rho))$. We retain the notation $R_s(\beta, \rho)$ when the residual is empty. To create a claim from the base ρ means to record the pair $(\rho, F_s(\beta, \rho))$; its claimed clopen set is exactly $R_s(\beta, \rho)$.

For each parent α and **coordinate** x, the construction recursively enumerates a sequence of **attempt records**

$$r_{\alpha,x,n} = (\sigma_{\alpha,x,n}, v_{\alpha,x,n}, b_{\alpha,x,n}, \mathcal{E}_{\alpha,x,n}).$$

Here $\sigma_{\alpha,x,n} \in 2^{b_{\alpha,x,n}}$ is the **guard**, $v_{\alpha,x,n}$ is the **value**, and $\mathcal{E}_{\alpha,x,n}$ is the **witness package**. The package records the worker β, its exact current claim (ρ, F), and the finite $W_e^A[s]$-computations selected when the record is issued. The records for (α, x) are numbered consecutively. Record $n + 1$ is issued only after the current approximation to A is incompatible with the guard of every record of index at most n. Issued attempt records are permanent entries in the ordered-search program: no later rule deletes them.

For every oracle Z, define the **ordered search** functional by $\Theta^Z(\alpha, x) = v_{\alpha,x,n}$, where n is least such that $\sigma_{\alpha,x,n} \subset Z$; if there is no such record, the computation diverges. The properties of this search are proved in Lemma 21 below.

A record issued at stage t with guard $\sigma = A_t \upharpoonright b$ is **live** at stage $s \geq t$ if $A_s \upharpoonright b = \sigma$. It is **killed** when this equality first fails. The **least gap** at α at stage s is

$$\text{gap}_\alpha(s) = \min\{x : \text{there is no live record for } (\alpha, x) \text{ at stage } s\}.$$

Suppose that the stored witness package has maximum oracle use w. If $x > 0$, let b_{x-1} be the guard length of the current live record at $x - 1$. A new record at the least gap x is issued according to the **nested-guard rule**

$$b = \begin{cases} w + 1, & x = 0, \\ \max\{w, b_{x-1}\} + 1, & x > 0, \end{cases} \qquad \sigma = A_s \upharpoonright b, \qquad v = f_s(x) + 1. \tag{1}$$

The number b need not exceed the guard lengths of previously killed records at coordinate x. This bounded-retry feature is essential in the density argument.

An **assignment** associates an attempt record with one child and is created only while that record is live. If an assigned record is killed, its assignment link is retained provisionally until the pending injury is processed, unless an intervening initialization unassigns it. Since assignments are created only for live records, once the link of a killed record has been removed, it can never be created again. At any stage, at most one record is assigned to a child and a record is assigned to at most one child. We write $\text{coord}_s(\beta) = x$ when the record at coordinate x is assigned to β, and write $\text{coord}_s(\beta) \uparrow$ when no record is assigned to β.

Suppose that a record assigned to β has coordinate x, was issued at stage t, and has value $v = f_t(x)+1$. It is **certified** at stage s if $f_s(x) > v$.

Each attempt record carries a Boolean **success mark**, initially off. Case 1 below is the only action which turns this mark on. We call a record **successful** while its success mark is on. A successful record is **valid for** β **at stage** s if it records β as its worker, is live at s, and the claimed clopen set stored in its witness package is exactly $R_s(\beta, \rho)$ for its stored base ρ. Thus all finite evidence of a successful certification is stored in the attempt record itself.

We say that a parent α is **visited** when its local module is processed during the stage traversal specified below; a visit to α means such an execution of its module.

After pending injuries (to be defined) have been processed, a worker β is **satisfied** at a visit if it has no current claim and, for every $\rho \in 2^{k_\beta}$, either $R_s(\beta, \rho) = \varnothing$ or there is a record valid for β which stores

exactly this residual. We call β **eventually satisfied** if it is satisfied at every sufficiently late visit to its parent, after pending injuries have been processed.

The construction uses a one-stage **coordinate cooldown** on a record coordinate and a finite **pending α-injury** for each parent α. The latter records the finite set of attempt records in the α-module killed since the previous visit, the coordinates killed with them by the nested-guard rule, and the assignments carried by those records when they were killed. Their operational effects are specified in the construction below.

A pair (ρ, F) is **resolved at stage** s if $s \geq \max\{|\xi| : \xi \in F \cup \{\rho\}\}$.

At a visit to a parent α, after its pending injury has been processed, the **frontier** is the first child having no assigned record. If the frontier β_i has current claim (ρ, F) with claimed clopen set $C = R(\rho, F)$, then this claim is **expansionary at stage** s if, for every $\tau \in 2^{s-1}$ with $[\tau] \subseteq C$, there is a string $\nu_\tau \in W_e^A[s]$ properly extending Ψ^τ, where $e = |\alpha|$. We choose witnesses having least oracle use and let w be the maximum of these uses.

To **initialize** a node γ means to remove its current claim, unassign any record assigned to it, turn off the success mark of every attempt record whose stored worker is γ, clear every coordinate cooldown in the γ-module, and delete any pending γ-injury. To initialize a subtree means to initialize every node in that subtree.

3.2. **The construction.** At stage 0, put $\Psi^\lambda = \lambda$. There are no claims, records, assignments, or active success marks, and all cooldowns and pending injuries are empty.

Whenever a claim for a node η is created or removed, apply the **claim-change initialization rule**: initialize every γ with $\eta <_L \gamma$. (Lemma 20 below proves that the claimed clopen sets produced by this rule are pairwise disjoint and have total measure at most $1/2$.)

At every finite stage, only finitely many nodes carry state on which initialization acts. Thus an instruction to initialize a collection of nodes means to apply the preceding initialization operations to its finitely many relevant members; all other members of the collection are already in their default state.

A general stage $s > 0$ proceeds as follows. At the beginning of the stage, before the traversal, inspect the finitely many records issued before stage s. Every record which was live at stage $s-1$ and is not live at stage s is killed and is entered, together with its coordinate and its assignment at that moment, into the appropriate pending injury. The stage traversal then begins at the root. Whenever it reaches a parent α, first process any pending α-injury and determine the resulting frontier. Then apply the first applicable local case; that case carries out the local action and specifies the outcome of α. The traversal follows this outcome to the corresponding child and repeats the same procedure, for at most s levels. After the traversal ends, define the stage-s output table for Ψ. The following three subsections specify these steps.

3.2.1. *Injury processing.* When a change in A kills some live records in the α-module, add those killed records, together with their coordinates and assignment data, to the pending α-injury; if further deaths occur before the next visit, their data are added to the same finite pending injury. It is processed at the next visit to α, before any other local action. If the module is never visited again, its pending injury is never processed, and the module performs no further action.

When processing this injury, turn off the success marks of all killed records. Among the killed records, consider only those whose assignment links are still present when the injury is processed. If there is no such record, initialize no child. Otherwise, let $\beta_j = \alpha j$ be the least-indexed child to which one of these killed records is still assigned. Unassign every record currently assigned to $\beta_j, \beta_{j+1}, \ldots$, leave the current claim of β_j unchanged, and initialize every β_ℓ with $\ell > j$, together with its subtree. The proper extensions of β_j are not initialized, because its claim has not changed and hence neither have their higher-priority claim sets. In all cases, put every coordinate recorded in the pending injury—not only the coordinates of records whose assignment links are still present—in one-stage cooldown. Delete the pending injury after these actions have been performed. Clear the resulting cooldowns at the end of the stage.

If a least affected child β_j exists, it is the resulting frontier and its retained current claim is nonempty, so Case 3 cannot apply at this visit. Moreover, the current least gap is among the coordinates just put in cooldown, so Case 4 cannot apply either.

3.2.2. *The local cases.* Suppose α is visited at stage s. First process any pending α-injury and let $\beta_i = \alpha i$ be the resulting frontier. (Lemma 23 below proves that precisely the children $\beta_0, \ldots, \beta_{i-1}$ are assigned and that their coordinates are strictly increasing.) Apply the first case whose hypotheses hold; it specifies the outcome of α.

Case 1. If the record assigned to some β_j, $j < i$, is certified, select the least such j. Use its witness package to extend Ψ throughout the claimed clopen set of its stored claim, and turn on its success mark. Then remove that claim and unassign the record. Apply the claim-change initialization rule and set the outcome of α to j. (The legitimacy of the stored computations is proved in Lemma 24 below.)

Case 2. If the frontier β_i is satisfied, make no state change and set the outcome of α to i.

Case 3. Suppose β_i has no current claim and the current stage resolves every nonempty pair $(\rho, F_s(\beta_i, \rho))$, $\rho \in 2^{k_{\beta_i}}$. Choose the least ρ whose current residual is nonempty and for which there is no valid record storing that exact residual. Create the claim $(\rho, F_s(\beta_i, \rho))$, apply the claim-change initialization rule, and set the outcome of α to i.

Case 4. If the frontier has a current expansionary claim and no assignment, and if $x = \mathrm{gap}_\alpha(s)$ is not in coordinate cooldown, issue an attempt record at x according to (1), assign it to β_i, and set the outcome of α to i.

Case 5. In every remaining case, make no state change and set the outcome of α to i.

Since the first applicable case is used, exactly one case is executed at a visit. Only the frontier is tested for satisfaction; no child to its right is processed by the local module.

3.2.3. *Stage traversal and the functional Ψ.* At stage s, start at the root and follow the outcomes just defined for at most s levels. The resulting finite sequence δ_s is the **stage path**; equivalently, its initial segments are exactly the nodes visited at stage s.

We define the output tables recursively at the end of stage $s \geq 1$. Fix $\tau \in 2^s$, let $\bar{\tau} = \tau \upharpoonright (s-1)$, and put $q = \Psi^{\bar{\tau}}$. Define Ψ^τ as follows.

(i) Suppose $[\tau]$ lies inside the claimed clopen set of a claim removed by Case 1 at stage s, and the corresponding attempt record was issued at stage u. Let $\eta = \tau \upharpoonright (u-1)$. Using the unique stored witness ν_η, put $\Psi^\tau = \nu_\eta$.
(ii) Otherwise, if $[\tau]$ lies inside the claimed clopen set of a current claim, put $\Psi^\tau = q$.
(iii) In every other case, put $\Psi^\tau = q^\frown 0$.

3.3. **Verification.** We first prove the elementary properties of the construction.

Lemma 20. *At every point of the construction after the initialization required by the current action has been performed, the claimed clopen sets of distinct current claims are pairwise disjoint. Moreover, the union of all current claimed clopen sets has measure at most* $1/2$.

Proof. When a claim for β is created, its claimed clopen set is a current residual and is therefore disjoint from the claimed clopen set of every surviving node $\gamma <_L \beta$. The claim-change initialization rule removes the claims of every node γ with $\beta <_L \gamma$. Removing further claims cannot create an intersection, so the claimed clopen sets remain pairwise disjoint. The claimed clopen set of a β-claim is contained in a cone of measure 2^{-k_β}. Since $\beta \mapsto k_\beta$ is one-to-one and $k_\beta \geq 2$, the union of all current claimed clopen sets has measure at most $\sum_{n \geq 2} 2^{-n} = \frac{1}{2}$. □

Lemma 21. *For every node α the following hold.*

(i) $\Theta^A(\alpha, \cdot)$ is an A-partial recursive function.
(ii) For every oracle Z, the domain of $\Theta^Z(\alpha, \cdot)$ is an initial segment of ω.
(iii) If $N \notin \mathrm{dom}(\Theta^A(\alpha, \cdot))$, then stages at which coordinate N has no live record occur cofinally.
(iv) Killing a live record at x kills every currently live record at a coordinate greater than x.
(v) At every stage, the coordinates carrying live records form an initial segment of ω.

Proof. First note that there is at most one live record at each coordinate. A new record at (α, x) is issued only after the current approximation to A is incompatible with the guards of all earlier records there, and the monotonicity of the r.e. approximation prevents any killed record from becoming live again. So part (i) follows from the ordered search. For (ii), suppose that a record at $x > 0$ has a guard compatible with Z. At its issue stage, its guard strictly extended the guard of the live record at $x - 1$. That lower guard is also compatible with Z, so some record at $x - 1$ is accepted by the ordered search. Induction gives downward closure.

If $N \notin \mathrm{dom}(\Theta^A(\alpha, \cdot))$, every record at N has a guard incompatible with A and is eventually killed. After each such record is killed, either α is never visited again, in which case no later N-record is issued, or its pending injury survives until the next visit, when the coordinate cooldown prevents the issue of an N-record. The only remaining possibility is that the pending injury is deleted because α is initialized before its next visit. Every construction step that does this also initializes all children of α, either by

the claim-change initialization rule or by initializing a subtree. Hence, at the first later visit to α, no frontier child has a current claim, so Case 4 cannot issue an N-record at that visit. Thus stages with no live N-record occur cofinally, proving (iii). Part (iv) follows directly from the same strict extension relation. For (v), records are issued only at the current least gap, while the death of a live record kills every live record at a higher coordinate by (iv). Certification and unassignment do not change which records are live. These observations preserve the asserted initial-segment property at every stage. □

Lemma 22. *Suppose a claim (ρ, F), with claimed clopen set $C = R(\rho, F)$, is created at stage t. Then (ρ, F) is resolved at stage t. If an attempt record for this claim is issued at stage u, then $u > t$, and the level-$(u-1)$ cones contained in C form a finite partition of C.*

Proof. Case 3 creates (ρ, F) only after the current stage resolves it. At most one local case is applied at a visit, so Case 4 cannot issue a record for the new claim until a later visit, at a stage $u > t$. Hence $u - 1 \geq t$, and every level-$(u-1)$ cone is either contained in or disjoint from C. The cones contained in C therefore form the asserted finite partition. □

Lemma 23. *Immediately after any pending α-injury, if one exists, has been processed at a visit to α, there is a unique i such that precisely $\beta_0, \ldots, \beta_{i-1}$ are assigned records. Their record coordinates are strictly increasing, every assigned record is live, and every assigned record stores its assignee as its worker and stores exactly that child's current claim. In particular, every assigned child has a nonempty current claim. Every child β_j with $j > i$ is in its initialized state and is not satisfied.*

Proof. The statement holds initially. The death of a record does not by itself remove its assignment link. Before the pending injury is processed, such a provisional link can disappear only through initialization, since no further local action in the α-module occurs before injury processing and a killed record cannot be assigned again. Every such initialization either initializes α itself, in which case the pending α-injury is deleted, or removes an assigned suffix of the children of α and initializes that suffix. Thus a killed assignment whose link is no longer present requires no further local initialization.

Suppose now that some killed assignment link is still present, and let β_j be the least child carrying one. By the inductive strict increase of the assigned coordinates and Lemma 21(iv), every record still assigned to a child to the right of β_j is killed as well. Injury processing therefore removes a suffix of the current assignments, initializes every child strictly to the right of the new frontier, and leaves only live assigned records. If no killed assignment link remains, either no assigned record was killed or the affected suffix has already been removed and initialized. The claim-change initialization rule likewise initializes a suffix of the children in any fixed local module.

Cases 2, 3, and 5 make no assignment. In Case 4, Lemma 21(v) shows that the current least gap is greater than every assigned coordinate. Assigning that coordinate to the frontier therefore preserves both the initial-segment form and the strict increase of the assigned coordinates. Case 1 unassigns one child and initializes every child to its right, again leaving an initial segment.

An assignment is created only in Case 4. At that action the attempt record stores the frontier as its worker and stores exactly the frontier's nonempty current claim. Case 1 removes that claim and unassigns the record simultaneously, and initialization also unassigns every affected record. Injury processing leaves the current claim of the least affected child unchanged while removing its assignment, and initializes every affected child to its right. Thus every remaining assigned record stores its assignee and exactly that child's nonempty current claim. The same induction shows that every child strictly to the right of the frontier is initialized: the local module processes only the frontier, and whenever the frontier moves left, it initializes the entire suffix to its right. Such a child has no claim, no assignment, and no active success mark. By Lemma 20, its higher-priority claim set has measure at most $1/2$, so some base has a nonempty residual. Hence the child is not satisfied. □

Lemma 24. *If an assigned attempt record is live at stage s, every computation in its stored witness package is still valid in $W_e^A[s]$. If its guard is compatible with the final set A, all of the stored strings belong to W_e^A.*

Proof. The guard length is strictly greater than every oracle use in the witness package. Liveness gives $A_s \upharpoonright b = \sigma$, so all computations in the package are still valid at stage s. If $\sigma \subset A$, the same computations are permanent and give membership in W_e^A. □

Lemma 25. *For fixed α and x, only finitely many distinct attempt records at coordinate x ever become certified.*

Proof. Choose S such that $f_s(x) = f(x)$ for every $s \geq S$. Every record at coordinate x issued at or after stage S has value $f_s(x)+1 = f(x)+1$, and therefore can never satisfy the certification inequality. Only finitely many records at this coordinate can have been issued before stage S. Hence only finitely many distinct records at coordinate x ever become certified. □

Lemma 26. *The clauses defining Ψ^τ are unambiguous and monotone and are uniformly recursive. Hence $\Psi^X = \bigcup_s \Psi^{X\restriction s}$ defines a recursive Turing functional. Moreover, while a claim remains current, Ψ is frozen on its claimed clopen set.*

Proof. By Lemmas 20 and 22, every level-s cone is either contained in or disjoint from every claimed clopen set relevant at stage s. The claimed clopen sets removed by Case 1 at the same stage are pairwise disjoint, and the claim-change initialization rule removes the appropriate lower-priority state before traversal continues. Thus clause (i) has a unique witness package.

Suppose a claim (ρ, F) with claimed clopen set C is created at stage t and removed by Case 1 at stage $s > t$. Clause (ii) freezes the output on C from the end of stage t through stage $s-1$. If its attempt record was issued at stage u, then, for every relevant τ,

$$\Psi^{\tau\restriction(s-1)} = \Psi^{\tau\restriction(u-1)} \subsetneq \nu_{\tau\restriction(u-1)}.$$

Hence clause (i) extends the previous value and the recursion is monotone. A claim created inside a region removed by Case 1 earlier in the same stage is subject to clause (i) at that stage; its frozen value begins at the end of its creation stage, and no record for it is issued until a later visit. All finite stage data are recursive, so the tables are uniformly recursive. The final assertions follow. □

The existence of the true path and the verification of the requirements must be proved simultaneously. The following stability notion records exactly the induction hypothesis needed at one parent.

Definition 27. *A node α is a **stable true parent** if it is visited infinitely often and, after some stage, the following hold.*

- *(a) No node lexicographically to the left of α is visited, and α is never initialized again by any construction step or injury processing.*
- *(b) All claims belonging to nodes $\gamma <_L \alpha$ are fixed.*
- *(c) Unless $\alpha = \lambda$, its claim status as a child in its parent's module is fixed in the following sense: either α is permanently satisfied or it has one fixed current claim. In the latter case, attempt records may still be assigned to and unassigned from α without changing that claim.*

A child is **settled** if, from some stage onward, one fixed live record is assigned to it. Such a record is compatible with A.

Lemma 28. *Let α be a stable true parent. If no child of α is eventually satisfied, then not every child is settled.*

Proof. Suppose every child β_i is eventually assigned a fixed record with coordinate x_i and value v_i. The coordinates are distinct and hence unbounded. The compatible records at arbitrarily large coordinates, together with Lemma 21(ii), make $\Theta^A(\alpha,\cdot)$ total. By Lemma 19, choose i so large that $f(x_i) > \Theta^A(\alpha, x_i) = v_i$. Eventually $f_s(x_i) > v_i$. At the next visit to α, Case 1 selects the least child among $\beta_0, \ldots, \beta_i$ whose assigned record is certified; Case 1 precedes the satisfaction test. It unassigns that child's fixed record, a contradiction. □

Let i be the least index such that β_i is not settled. Every β_j with $j < i$ is settled. By Lemma 23, the assigned children form an initial segment, so outcomes $j < i$ eventually cease. Their subtrees receive no later visits, their claims remain fixed, and no later construction step to the left initializes β_i. Thus the higher-priority claim set of β_i is fixed from some stage onward.

Lemma 29. *Assume that α is a stable true parent, no child of α is eventually satisfied, and i is the least index of a nonsettled child. Then $\Theta^A(\alpha,\cdot)$ is not total.*

Proof. Suppose it is total. Final A-compatible records at arbitrarily large coordinates are issued arbitrarily late. After the stabilization of the children to the left of i, every such sufficiently late record is issued at a visit whose frontier is some β_j with $j \geq i$.

If $j = i$, the record assigned to β_i is compatible with A. If $j > i$, Lemma 23 says that a lower-coordinate record is assigned to β_i at that moment, and the nested-guard rule implies that this record is compatible with A as well. It cannot be killed, and no later initialization from the left can unassign it. If it were never certified, β_i would be settled, contrary to the choice of i. It is therefore eventually

certified. Case 1 processes it and makes it successful. Because the record is compatible with A and the higher-priority claim set of β_i is fixed, and because β_i is never initialized again, its success mark remains on and the record remains valid for β_i permanently.

Case 1 must make infinitely many such permanent records successful for β_i. Otherwise, fix a stage after the last one is processed and after all the stabilization described above. By totality, a final compatible record is issued later at a visit whose frontier is some β_j with $j \geq i$. At its issue stage, a compatible record is assigned to β_i: it is the newly issued record if $j = i$, and it is a lower-coordinate record if $j > i$. This record cannot be killed, and no initialization from the left can unassign it. Hence it must eventually be certified; otherwise β_i would be settled. Case 1 then makes another permanent record successful, a contradiction. A temporarily satisfied state or a permanently nonexpansionary claim cannot occur throughout this argument: in the first case β_i would be eventually satisfied, and in the second case only finitely many coordinates of the total ordered search could receive records.

Since the higher-priority claim set of β_i is fixed, each permanently valid successful record produced in this way corresponds to a base for which there was previously no valid record. After this stabilization, the residual at each base is fixed, so a permanently valid record for that residual prevents Case 3 from ever selecting the same base again. The records therefore correspond to distinct bases. There are only $2^{k_{\beta_i}}$ bases, contradicting the existence of infinitely many such records. □

Put $N = \min\big(\omega \setminus \mathrm{dom}(\Theta^A(\alpha, \cdot))\big)$. Every coordinate below N has a permanent compatible record after some stage, whereas every record at coordinate N is eventually killed.

Lemma 30. *Under the hypotheses of Lemma 29, the least nonsettled child β_i eventually keeps one fixed current claim (ρ, F) permanently; its claimed clopen set is a nonempty residual.*

Proof. If the settled child β_j, $j < i$, has permanent assigned coordinate x_j, then its record is compatible with A, so $x_j \in \mathrm{dom}(\Theta^A(\alpha, \cdot))$. By the definition of N, we therefore have $x_j < N$.

Fix a stage after all of the following conditions hold.

- The children β_j, $j < i$, and the higher-priority claim set of β_i have stabilized.
- Every fixed finite prefix-free representation of a residual $R_s(\beta_i, \rho)$, $\rho \in 2^{k_{\beta_i}}$, is resolved by the current stage.
- For every $m < N$, the final least A-compatible record selected by the ordered search has appeared, so m is never again a gap.
- No record at a coordinate below N will again be assigned to β_i.
- Every record at coordinate N that will ever become certified has already reached its first certification stage.

The resolution condition is legitimate because the higher-priority claim set is fixed and there are only finitely many bases. The fourth item is legitimate: if a final record at some $m < N$ is assigned to β_i, then it either remains assigned permanently, making β_i settled, or it is eventually certified and unassigned. There are only finitely many $m < N$. The fifth item follows from Lemma 25.

By Lemma 21(iii), choose a sufficiently late stage at which there is no live record at coordinate N. By Lemma 21(v), there is then no live record at a higher coordinate. If the gap results from a pending injury, process it at the next visit to α; injury processing unassigns the affected suffix and places coordinate N in one-stage cooldown, so no record can be issued at N at that visit. If there is no pending injury, no record can be issued before the next visit to α. In either case, N remains the least gap until the next permitted visit.

Call a later visit *permitted* if, after pending injury processing, N is not in cooldown. Such visits occur arbitrarily late: cooldowns last only one stage, and every record at N is eventually killed. Moreover, if β_i were satisfied at every sufficiently late permitted visit, it would be eventually satisfied. Indeed, after a required success mark is lost, no replacement can become successful until Case 3 has first created a claim at one visit; that claim is still present at the next permitted visit. Thus every sufficiently late failure of satisfaction produces a failure at a permitted visit as well.

At every permitted visit before a new record at N is issued, the settled children β_j, $j < i$, retain their assigned coordinates below N. The child β_i has no assignment below N, and the absence of a live record at N excludes every higher-coordinate assignment after pending injuries have been processed. Hence β_i is the frontier.

Now consider the first permitted later visit at which β_i is not satisfied. Such a visit exists unless β_i is eventually satisfied, contrary to the hypothesis. If it already has a current claim, denote that claim by (ρ, F). Otherwise, the resolution condition above ensures that Case 3 creates a claim (ρ, F) for the least base whose residual is nonempty and has no valid record.

No settled child β_j, $j < i$, can have its fixed record certified at this late stage, since the next visit to α would then unassign it. Thus no later Case 1 action to the left removes (ρ, F); the claim can be removed only by a certification of a record assigned to β_i.

If there is no permitted later visit at which this claim is expansionary, then (ρ, F) is already permanent. Otherwise, at the first such visit Case 4 issues a record at the current gap N and assigns it to β_i. By our choice of stage, this record cannot be certified. When the record is killed, the next injury-processing visit unassigns it and leaves the claim (ρ, F) unchanged. The least gap is again N, and the same reasoning applies to each later record issued at N. Because pending injuries are processed before Case 4, no child to the right of β_i can receive a record at N. Hence the claim (ρ, F) is never removed and is permanent. □

Lemma 31. *Let α be a stable true parent assigned to Q_e. If W_e^A is dense, then some child of α is eventually satisfied.*

Proof. Suppose no child is eventually satisfied. Let β_i, N, and the permanent claim (ρ, F) be obtained from Lemmas 28–30, and put $C = R(\rho, F)$. By Lemma 26, Ψ is frozen on C.

Choose a finite level L which resolves (ρ, F) and is at least the stage at which the permanent claim was created and its output became frozen. For every $\tau \in 2^L$ with $[\tau] \subseteq C$, choose by density a string $\nu_\tau \in W_e^A$ properly extending Ψ^τ. There are only finitely many such strings. Choose actual A-oracle enumeration computations witnessing their membership and let w be the maximum of their uses. If $N > 0$, let b_{N-1} be the guard length of the final least A-compatible record selected by the ordered search at $N-1$; if $N = 0$, put $b_{-1} = 0$. Set $B = \max\{w, b_{N-1}\}+1$. Choose $s_0 \geq L+1$, after all the stabilization stages used above, such that $A_s \upharpoonright B = A \upharpoonright B$ for every $s \geq s_0$, and such that all the chosen witnesses have appeared by stage s_0.

By Lemma 21(iii), choose a stage after s_0 at which there is no live record at N. Process at the next visit to α any pending injury caused by the loss of the record at N. If such an injury is processed, the one-stage cooldown prevents the issue of a new record at N and hence any new assignment at that visit. If there is no pending injury, no record can be issued before the next visit to α. Let r be the first subsequent visit at which the resulting one-stage cooldown, if any, has expired; in the no-injury case, take r to be the next visit. Since only this local module can issue a record at N, N remains the least gap until the local cases are considered at r.

At that point every settled child β_j, $j < i$, retains its assigned coordinate below N, while β_i has no assignment below N. Lemma 21(v) and the absence of a live record at N exclude every live record at a higher coordinate; after pending injuries have been processed, the frontier index is therefore i. No record assigned to a settled β_j, $j < i$, can be certified at this late stage, and the permanent claim prevents β_i from being satisfied.

For every $\eta \in 2^{r-1}$ with $[\eta] \subseteq C$, put $\tau = \eta \upharpoonright L$. The recursive definition of Ψ and the freezing on C give $\Psi^\eta = \Psi^\tau$. Hence the previously chosen ν_τ is a valid witness for η. The claim is expansionary. Case 1 does not apply because no record assigned to a settled β_j, $j < i$, is certified at this late stage; Case 2 does not apply because β_i has the claim (ρ, F); and Case 3 does not apply because that claim is already present. Therefore Case 4 applies. Since the construction chooses least-use witnesses, every selected witness has use at most w, and the new guard length in (1) is at most B. Its guard is therefore an initial segment of the true A. This record witnesses $N \in \mathrm{dom}(\Theta^A(\alpha, \cdot))$, contradicting the definition of N. □

A child $\beta_i = \alpha i$ is a **stable true child** of α if it is visited infinitely often and, after some stage, no node lexicographically to its left is visited, β_i is never initialized again, every claim belonging to a node $\gamma <_L \beta_i$ is fixed, and its claim status in the α-module is fixed in the following sense: it is either permanently satisfied or has one fixed current claim. In the latter case, its attempt-record assignment need not stabilize.

Lemma 32. *Every stable true parent α has a stable true child.*

Proof. First suppose some child is eventually satisfied. By Lemma 23, every assigned child has a nonempty current claim and is therefore not satisfied, while every child strictly to the right of the frontier is initialized and not satisfied. Consequently, the satisfied child is unique and is the frontier. Work after the stage from which this frontier is satisfied at every visit to α. No record assigned to an earlier child can subsequently be killed or certified: the resulting injury processing or claim removal would initialize the frontier and turn off all its success marks, so it would fail to be satisfied at the next visit. For the same reason, no claim in its higher-priority claim set can change, and no successful record needed to witness its satisfaction can be killed or otherwise become invalid. Thus the frontier index, its residuals, and its witnessing successful records are all permanent. Case 2 applies at every late visit, and

this child is visited at every late visit to α. No smaller child is visited later. Its satisfied status and empty claim are permanent, and it is never initialized again. Claims in smaller sibling subtrees remain fixed because injury processing is local and those subtrees receive no later visits; claims further to the left were already fixed for the stable true parent. Together with the stability of α, the absence of smaller outcomes shows that no node lexicographically to the left of this frontier is visited later. Hence this frontier is a stable true child.

Now suppose no child is eventually satisfied. Let i be the least index of a nonsettled child. Every $j < i$ is eventually assigned permanently, so all late frontier outcomes are at least i, and the subtrees of smaller children receive no more visits. By Lemma 30, β_i has a permanent claim. If that claim is eventually never expansionary, β_i remains the frontier at every late visit. Otherwise it repeatedly has an A-incompatible record at N assigned to it. Every such record is eventually killed, and after the corresponding injury is processed β_i is again the frontier; the visit at which the next record at N is issued has outcome i. Thus outcome i occurs infinitely often, while no smaller outcome occurs late. No later construction step to the left initializes β_i, and both its current claim and all claims to its left are fixed. The stability of α and the absence of smaller outcomes also imply that no node lexicographically to the left of β_i is visited later. Therefore β_i is a stable true child. □

The root λ is a stable true parent by convention. Iterating Lemma 32 constructs the **true path**, whose nodes are stable true parents.

Lemma 33. *Let α be the stable true-path node assigned to Q_e. If one of its children is eventually satisfied, then Q_e is satisfied.*

Proof. Let β be the eventually satisfied child. Its higher-priority claim set is eventually fixed, since the creation or removal of a claim in that set would initialize β and turn off all of its success marks. Work after this stabilization and after β is satisfied at every visit. For each fixed nonempty residual, there is then at most one valid successful record: while such a record is valid, Case 3 cannot select its base again. If its guard were incompatible with A, the record would eventually be killed and its success mark would be turned off at the next visit. No replacement can become successful at that visit, since Case 3 must first create the corresponding claim and only one case is applied. Thus β would not be satisfied at that visit, a contradiction. Hence every successful record witnessing the late satisfaction of β is compatible with A and permanently valid.

Write $\mathcal{H}(\beta)$ and $R(\beta, \rho)$ for the stabilized values of $\mathcal{H}_s(\beta)$ and $R_s(\beta, \rho)$. Fix X such that Ψ^X is total, and put $\rho = X \upharpoonright k_\beta$. If X belongs to the fixed union $\mathcal{H}(\beta)$ of higher-priority claimed clopen sets, then one of the corresponding fixed claims freezes Ψ along X by Lemma 26; Lemma 20 prevents a later lower-priority claim from changing the output on that region. This contradicts totality. Hence $X \in R(\beta, \rho)$.

By eventual satisfaction, there is a permanently valid successful record for this exact fixed residual. When Case 1 made that record successful, the construction extended Ψ throughout the residual to strings enumerated in $W_e^A[s]$. Lemma 24 and compatibility of the guard with A make these enumerations correct in W_e^A. Therefore some initial segment of Ψ^X belongs to W_e^A. This is Q_e. □

Proof of Theorem 6. By Lemma 26, the stage construction defines the recursive functional Ψ. The same proof as Lemma 17 gives M, this time we use Lemma 20. Iterating Lemma 32 gives a stable true-path node α of every length. If α is assigned to Q_e and W_e^A is dense, Lemma 31 gives an eventually satisfied child, and Lemma 33 verifies Q_e. Thus every Q_e holds. The discussion in Subsection 2.1 now shows that A is AEWG. □

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